

Conditional Statements

Choosing which branch runs

Gregory S. DeLozier, PhD

Breaking Changes

`if` and `else` become keywords.

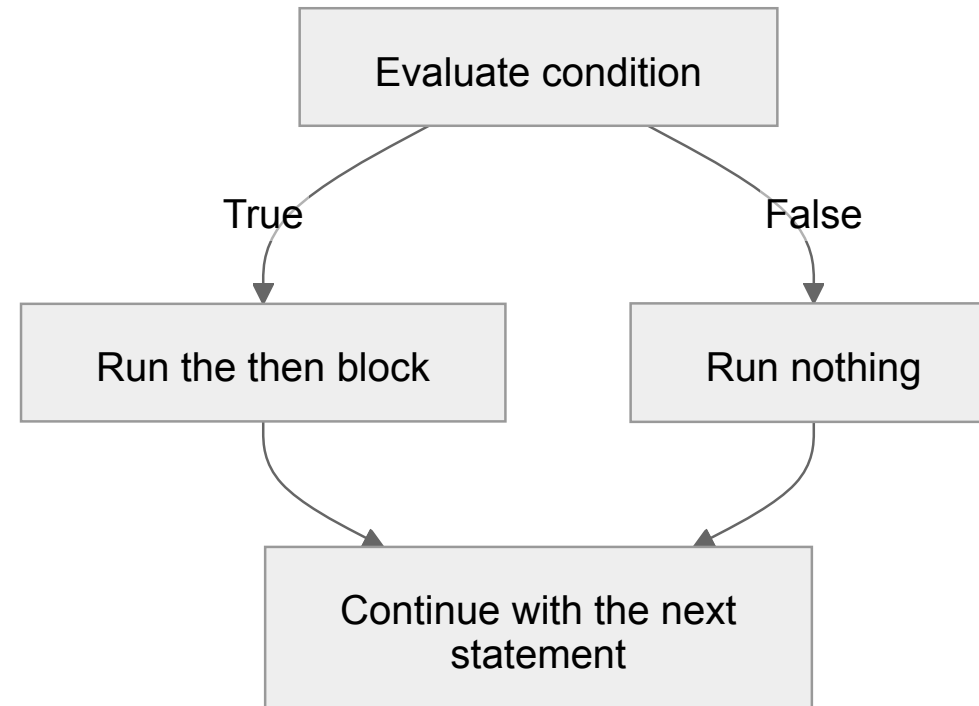
`{` and `}` are now reserved for blocks.

Only One Branch Runs

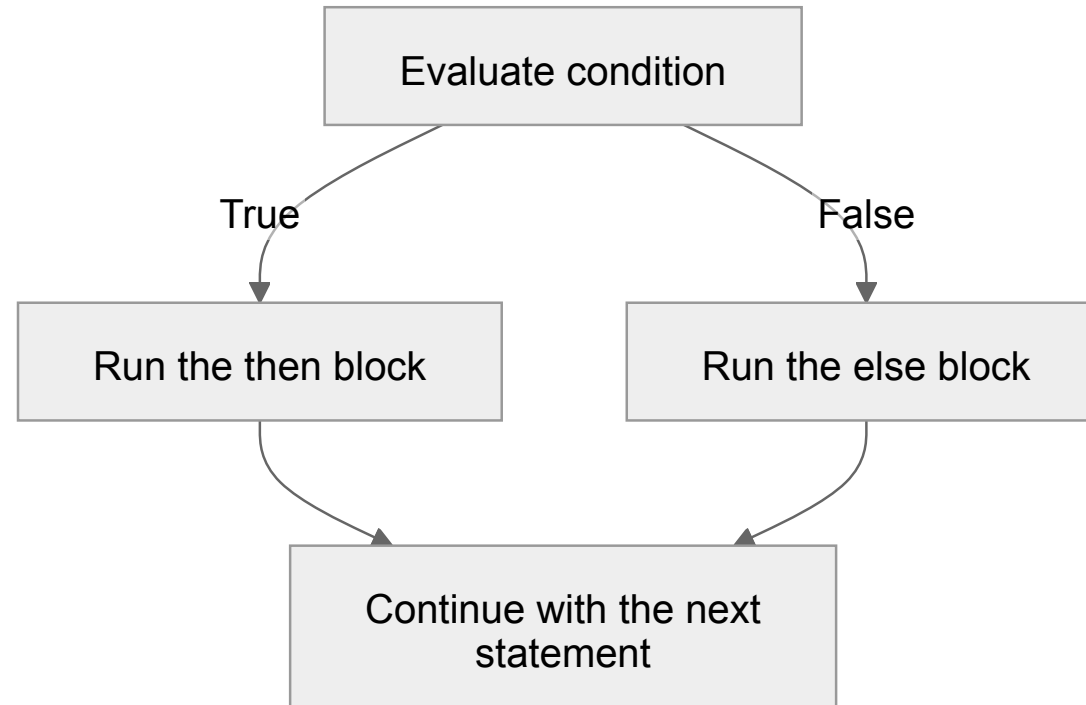
```
if (temperature > 90) {  
    print("Too hot");  
} else {  
    print("Fine");  
}
```

The condition must be a boolean.

If, No Else



If, With Else



Exactly one branch runs, either way.

The Unselected Branch Is Never Evaluated

```
if (false) {  
    stuck = exit(9) + number(input());  
} else {  
    print("only this branch runs");  
}
```

Not skipped output. Skipped evaluation.

Every Branch Is a Block

```
if_statement ::= "if" "(" expression ")" block [ "else" block ]  
block ::= "{" statement_list "}"
```

A bare statement is never a valid branch.

Why Require Braces Everywhere

```
if (a) if (b) x=1; else y=2;
```

Which `if` does the `else` belong to?

No “else if” Shortcut

```
if (grade >= 90) {  
    letter = "A";  
} else {  
    if (grade >= 80) {  
        letter = "B";  
    } else {  
        letter = "C or lower";  
    };  
};
```

Parsing a Block

```
def parse_block(tokens):  
    tokens = require(tokens, "{", "Expected '{' to start a block")  
    statements, tokens = parse_statement_list(tokens)  
    tokens = require(tokens, "}", "Expected '}' to close a block")  
    return statements, tokens
```

A Second Caller for `statement_list`

```
if tokens[0]["tag"] in (None, "}"):
    break
```

A trailing semicolon before `}` is now allowed too.

Semicolons Around a Conditional

```
if (true) { x=1 };  
y=2;
```

A semicolon after }, none before else.

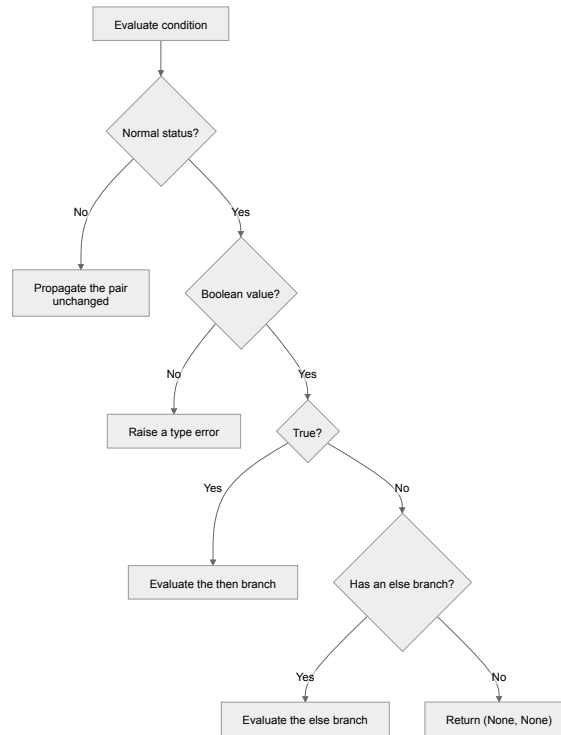
Parsing an If Statement

```
def parse_if_statement(tokens):
    tokens = require(tokens, "if", "Expected 'if'")
    tokens = require(tokens, "(", "Expected '(' after 'if'")
    condition, tokens = parse_expression(tokens)
    tokens = require(tokens, ")", "Expected ')' after if condition")
    then_block, tokens = parse_block(tokens)
    else_block = None
    if tokens[0]["tag"] == "else":
        else_block, tokens = parse_block(tokens[1:])
    return {"tag": "if", "condition": condition, "then": then_block,
            "else": else_block}, tokens
```

Evaluating a Conditional

```
if ast["tag"] == "if":
    condition, status = evaluate(ast["condition"], environment)
    if status is not None:
        return condition, status
    if type(condition) is not bool:
        raise TypeError("if condition must be a boolean expression")
    if condition:
        return evaluate(ast["then"], environment)
    if ast["else"] is not None:
        return evaluate(ast["else"], environment)
    return None, None
```

The Full Decision



Status Still Escapes Nested Blocks

```
x = 1;
if (true) {
    if (true) {
        exit(7);
    };
};
x = 2;
```

Result: (7, "exit"). x stays 1.

Testing the Contract

Single- and two-branch conditionals, nested conditionals, multi-statement blocks.

Unselected branch never touches input, exit, or assert.

Status escapes nested blocks and stops later statements.

Parser rejects unbraced bodies; accepts braced single-statement and nested blocks.

Questions for Thought

Why must the unselected branch be skipped entirely, not evaluated and discarded?

What would break if `if` consumed a status from its own selected branch?

What would adding an “else if” shortcut change about how `else` attaches?

If blocks later introduced their own scope, what would have to change here?